(KTXFI2EBNF) Physics II. Homework Solutions

October 27, 2025

Constants used

$$\begin{aligned} h &= 6.62607015 \cdot 10^{-34}\ \mathrm{J \cdot s} \\ c &= 2.99792458 \cdot 10^{8}\ \mathrm{m/s} \\ e &= 1.602176634 \cdot 10^{-19}\ \mathrm{C} \\ m_e &= 9.1 \cdot 10^{-31}\ \mathrm{kg} \quad \text{(value given in the homework)} \end{aligned}$$

Problems and solutions

1. An electron has a kinetic energy of 2.5 eV.

   (a) What is its velocity?

   Convert energy to joules $E_{kin} = 2.5\,\mathrm{eV} = 2.5 \cdot e$ and use $E_{kin} = \frac{1}{2} m_e v^2$.

   $$v = \sqrt{\frac{2E_{kin}}{m_e}} \approx 9.382520 \cdot 10^{5}\,\mathrm{m/s}$$

   (b) What is its momentum?

   $$p = m_e v \approx 8.538093 \cdot 10^{-25}\,\mathrm{kg\ m/s}$$

   (c) What is the wavelength of the associated de Broglie wave?

   $$\lambda = h/p \approx 7.760597 \cdot 10^{-10}\,\mathrm{m}$$

2. A metal surface is illuminated with light of wavelength $2.5 \cdot 10^{-7}\,\mathrm{m}$. What is the velocity of the emitted electrons if the photoelectric effect starts at light with a wavelength of $2.77 \cdot 10^{-7}\,\mathrm{m}$? ($m_e = 9.1 \cdot 10^{-31}\,\mathrm{kg}$)

   The maximum kinetic energy is $E_{kin} = hc/\lambda - hc/\lambda_0$ (if positive), hence

   $$\begin{aligned} E_{kin} &\approx 7.744987 \cdot 10^{-20}\,\mathrm{J}, \\ v &= \sqrt{\frac{2E_{kin}}{m_e}} \approx 4.125767 \cdot 10^{5}\,\mathrm{m/s}. \end{aligned}$$

3. A photon ejects an electron with a maximum kinetic energy of 0.64 eV from a metal for which the work function is 3.74 eV.

   (a) What is the energy of the photon in electronvolts?

   $$E_{\mathrm{ph}} = K + \phi = 4.38\ \mathrm{eV}.$$

   (b) What is the wavelength of the applied ultraviolet radiation?

   $$\lambda = hc/E_{\mathrm{ph}} \approx 2.830689 \cdot 10^{-7}\,\mathrm{m}.$$

4. In a Compton scattering experiment, X-rays with a wavelength of 0.104 nm are used. ($m_e = 9.1 \cdot 10^{-31}$ kg)
   (a) At what scattering angle does the wavelength of the radiation increase by 1%?
   The Compton formula $\Delta\lambda = \lambda_C(1 - \cos\theta)$ with $\lambda_C = h/(m_e c)$ gives $\theta \approx 55.1237°$.
   (b) At what angle does the wavelength become 0.05% larger?
   $\theta \approx 11.8774°$.
5. X-rays are scattered by electrons. $\lambda_0 = 10^{-11}$ m. The magnitude of the wavelength change is $2.6 \cdot 10^{-12}$ m.
   (a) What is the scattering angle of the photons?

   Using $\Delta\lambda = \lambda_C(1 - \cos\theta)$, $\theta \approx 94.0417°$.
   (b) By how much did the photon energy change during the process? ($m_e = 9.1 \cdot 10^{-31}$ kg)

   Photon energies before and after are $E_i = hc/\lambda_0$ and $E_f = \frac{hc}{\lambda_0 + \Delta\lambda}$, so the change is

$$\Delta E = E_i - E_f \approx 4.099015 \cdot 10^{-15}\,\text{J} = 25584.040947\,\text{eV}.$$

Dr. Gabor FACSKO
facsko.gabor@uni-obuda.hu